



Evolutionary Journeys

01

Origins of Migration





First Movers

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Early Human Observations

The earliest evidence of humans observing animal migration dates back to the Stone Age, with rock art images portraying animals moving across the African savannah, produced by nomadic hunter-gatherers around 20,000 years ago.

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Ancient Theories

Aristotle, the ancient Greek philosopher, was one of the first to theorize about animal migration. He speculated that some birds transformed into different species during winter, a belief that persisted until modern times.

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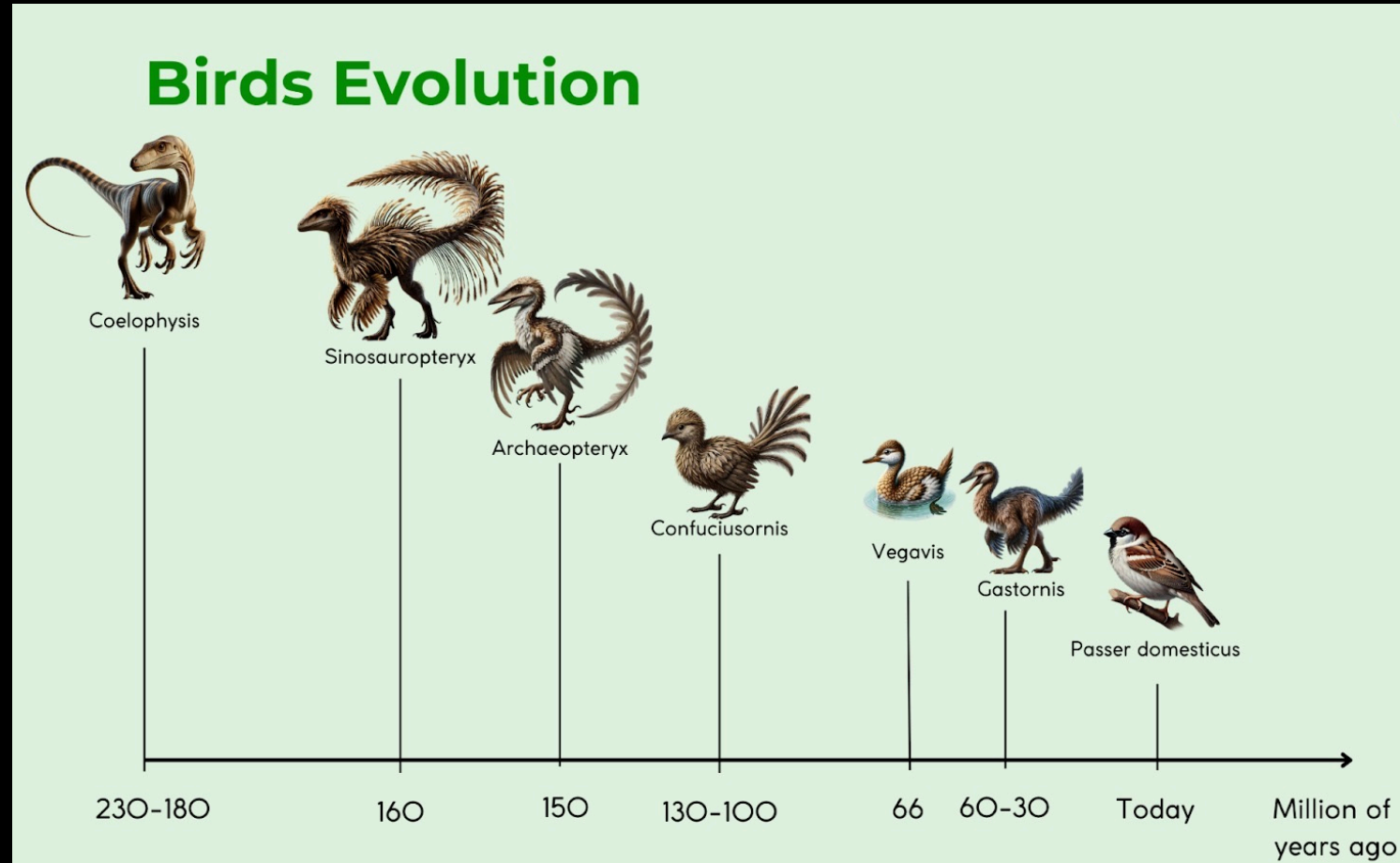
Aristotle's theory

Aristotle carefully watched the timing of arrivals and departures. Greece is a summer home to birds like redstarts (which migrate to Africa in winter) and a winter home to birds like robins (which come from north Europe). When he noticed robins arriving just as redstarts disappeared, he connected the dots as if birds morphed rather than migrated.



► Ancient Traces in Stone and Genes

Fossil records suggest that early avian migration began at least 250,000 years ago. Genetic studies reveal that migration patterns are deeply rooted in an animal's genetic makeup, with specific genes influencing behaviors like timing, route selection, and navigation, showing that migration is an ancient trait.



► Why Travel Evolved

Resource Tracking

Migration evolved as a strategy for animals to track seasonal resources, ensuring they have access to food and suitable habitats throughout the year, enhancing their survival and reproductive success.

Predator Escape

Moving to different locations helps animals avoid predators and unfavorable conditions, providing a significant survival advantage and driving the evolution of migratory behaviors.



Reproductive Advantage

Migrating to specific breeding grounds allows animals to reproduce in areas with fewer competitors and predators, increasing the chances of offspring survival and contributing to the spread of advantageous traits.

Natural Selection

Over time, natural selection refined migratory traits such as timing, fuel use, and navigation, embedding these behaviors in the genomes of migratory species and ensuring their continuation.



02

Simplest Migrants



► Tiny Brains, Long Voyages

Minimalist Migrants

Simple organisms like black-bean aphids and monarch butterflies undertake multi-step migrations. Despite their small size and limited nervous systems, they achieve remarkable journeys by responding to basic environmental cues.



▶ Innate Programs



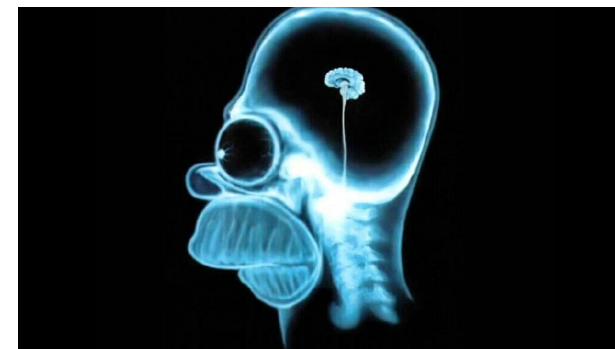
Inherited Navigation

Monarch butterflies are born with an innate ability to navigate using the sun's position and an internal circadian clock, allowing them to undertake long migrations without learning from their parents.



Aphid Alternation

Black-bean aphids alternate between sexual and asexual generations, each with specific migratory roles. This innate alternation ensures that the species can migrate effectively without complex learning.



Efficient Survival

These simple migratory programs are highly efficient, allowing species to survive and thrive in changing environments without the need for complex cognitive abilities.

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Relay across Generations



► Monarchs' Generational Relay



Multi-Generational Journey

Monarch butterflies undertake a multi-generational migration, with the initial generation living up to eight months and flying from Canada and the U.S. to Mexico. Subsequent generations complete the return trip, each contributing to the journey.

Inherited Instinct

Despite no single butterfly completing the full round trip, each generation inherits the instinct to migrate, using environmental cues and genetic programming to navigate to their destination.

► How Offspring Know the Way

Antennal Compass

Monarch butterflies rely on their antennae to sense the position of the sun and use an internal circadian clock to maintain their migratory heading, ensuring they stay on course.

Magnetic Field

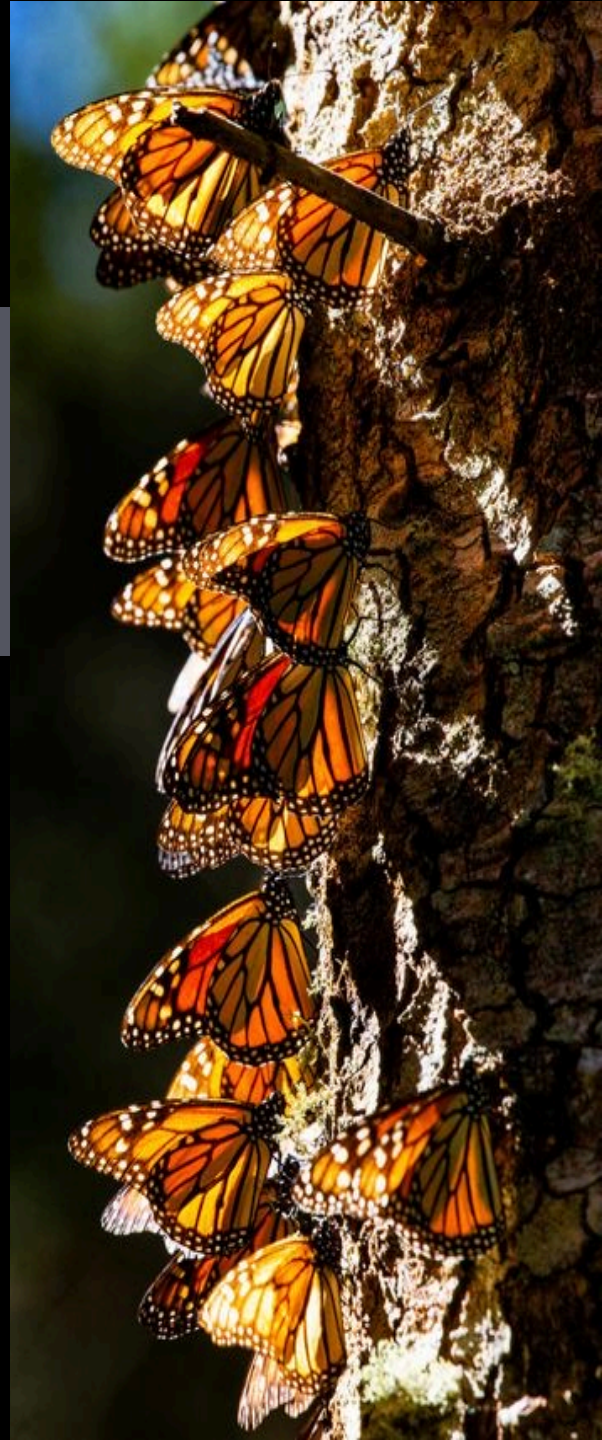
In addition to solar navigation, monarchs may also use the Earth's magnetic field to guide their migration, providing an additional layer of navigational precision.

Genetic Expression

Genes controlling migratory traits are differentially expressed between generations, switching on longevity and flight capabilities as needed, demonstrating the genetic basis of migration.

Environmental Cues

Environmental factors like temperature and daylight hours trigger the migratory behavior in monarchs, ensuring that each generation is ready to undertake its part of the journey.



04

Broader Evolutionary Lessons





From Insects to Whales

Diverse Migrants

Migration is a widespread phenomenon across the animal kingdom, from tiny insects like monarch butterflies to massive mammals like humpback whales. Each species has evolved unique strategies to undertake these journeys.



Shared Traits

Despite the diversity, common traits like fat loading, navigation modules, and hormonal triggers are shared among migratory species, highlighting the convergent evolution of migration.





Migration as Innovation Driver

Speciation Catalyst

Migration accelerates speciation by isolating populations along different migratory routes, leading to genetic divergence and the emergence of new species.

Ecological Feedback

The constant feedback between genome, physiology, and environment in migratory species makes them ideal models for studying evolutionary processes in real-time.

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Gene Flow

Migratory species can also spread advantageous alleles across geographically distanced populations, promoting genetic diversity and rapid adaptation to changing environments.

Conservation Importance

Understanding the evolutionary dynamics of migration is crucial for conservation efforts, as it helps identify key threats and develop strategies to protect migratory species and their habitats.





Next Steps in Migration Science



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Genomic Insights

Combining genomics with tracking technologies will reveal how flexible migration is and how species adapt to changing environments, providing deeper insights into evolutionary processes.

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Citizen Science

Engaging citizen scientists in tagging and tracking efforts can generate vast amounts of data, helping researchers monitor migratory patterns and identify emerging trends.

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Conservation Strategies

Developing conservation strategies that account for migratory behaviors, such as creating protected corridors and managing habitat connectivity, will be crucial for preserving these evolutionary journeys.





THANK YOU